Analyzing Breakdown by Strike Charts

Positional breakdown by strike

Visualizing net positions broken down by strike is a powerful tool. This method allows us to quickly understand the directional bias of a position and the potential hedging requirements or responses of market makers as the underlying price approaches certain levels.

Options contracts come in two types: calls and puts, and each can either be bought or sold. With thousands of transactions occurring for each contract daily, tracking them visually can be an extremely arduous task. Fortunately, computers can simplify this process.

By analyzing the flow, one can understand the momentary bias and infer the direction of the market. However, this approach may often lag behind the actual price action. Understanding the global picture by knowing the net position of customers can provide a significant edge, revealing the overall supply and demand imbalance for each contract.

This imbalance often acts as a blueprint, shaping the price action for the session. But if markets are efficient, how can customers be net long or short positions? This is where different market participants come into play. Market makers, for example, provide liquidity to the market, allowing traders or institutions to execute trades quickly and at reasonable prices. They fill the imbalance by taking the opposite side of customers' trades.

Reading these charts allows us to quickly identify key areas of interest for a trading day. Here, we will dive into the net customer position, delta exposure (DEX), and market makers' net gamma exposure (GEX).

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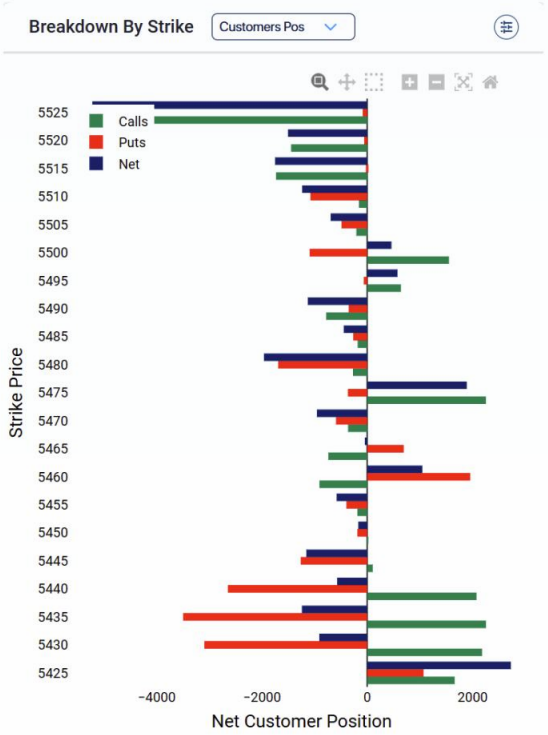
Net customer position (Customers Pos)

Let's start off with this table.

Net Customer Position	Position	Side	Color of the Bar	Bias
Net Long **Calls**	Positive (+)	Right	**Green**	Bullish
Net Long **Puts**	Positive (+)	Right	**Red**	Bearish
Net Short **Calls**	Negative (-)	Left	**Green**	Bearish
Net Short **Puts**	Negative (-)	Left	**Red**	Bullish



Customers Pos Potential Scenarios

Selling OTM Calls indicates a bearish outlook on the underlying asset. Customers who sell OTM calls believe that the asset's price will remain below the strike price, allowing them to keep the premium received from selling the call without having to deliver the underlying asset. This strategy benefits from a stable or declining market.

Selling OTM Puts signals a bullish outlook on the underlying asset. Customers who sell OTM puts expect the asset's price to stay above the strike price, allowing them to keep the premium received from selling the put without having to buy the underlying asset at the strike price. This strategy benefits from a stable or rising market.

Buying OTM Calls provides upside exposure. When customers buy calls, they are anticipating that the underlying asset's price will rise before the option expires. This strategy benefits from bullish market movements, allowing the call buyer to potentially profit from significant upward price swings. As the underlying price increases, the call option's value increases, offering leveraged gains with limited initial investment.

Buying OTM Puts provides downside exposure. When customers buy puts, they are expecting the underlying asset's price to drop before expiration. This strategy benefits from bearish market movements, protecting the put buyer from declines in the underlying asset's price. As the underlying price decreases, the put option's value increases, providing a way to profit from or hedge against significant downward price movements.

Deep In-the-Money (ITM) Options are often part of broader strategies. These options have strike prices significantly below the current spot price for calls, or significantly above the spot price for puts. Deep ITM options have deltas close to 1 for calls and -1 for puts, making them behave similarly to the underlying asset with less capital outlay. They are frequently used in strategies like covered calls, protective puts, or as part of spreads, where the investor aims to maximize leverage while minimizing risk. Due to their high intrinsic value and low time value, deep ITM options provide a way to gain substantial exposure to the underlying asset's price movement with relatively lower volatility.

Positions might change over time, but they provide a good idea of the overall sentiment*. ***Added precision will be covered in our intraday data subscription.***

An interesting aspect is the zero days to expiration (0DTE) options, where large net positions are often held for most of the day. Current data suggests that intraday volume for 0DTE options is generally balanced between buys and sells. Large positions are often held until expiration, providing key insights into market sentiment and potential price movements.

Customer Delta Exposure (Customers DEX)

Net Customer Position	Position	Side	Color of the Bar	Bias
Net Long **Calls**	Positive (+)*	Right	**Green**	Bullish
Net Long **Puts**	Negative (-)*	Left	**Red**	Bearish
Net Short **Calls**	Negative (-)	Left	**Green**	Bearish
Net Short **Puts**	Positive (+)	Right	**Red**	Bullish

**A positive delta infers a bullish outlook*
**A negative delta infers a bearish outlook*

Delta Characteristics

The delta of an option contract is dynamic:

- The delta of a single call ranges from 0 to 1.
- The delta of a single put ranges from -1 to 0.
- In-the-money options have a greater weight, with the delta of a single call approaching 1 and a single put approaching -1.

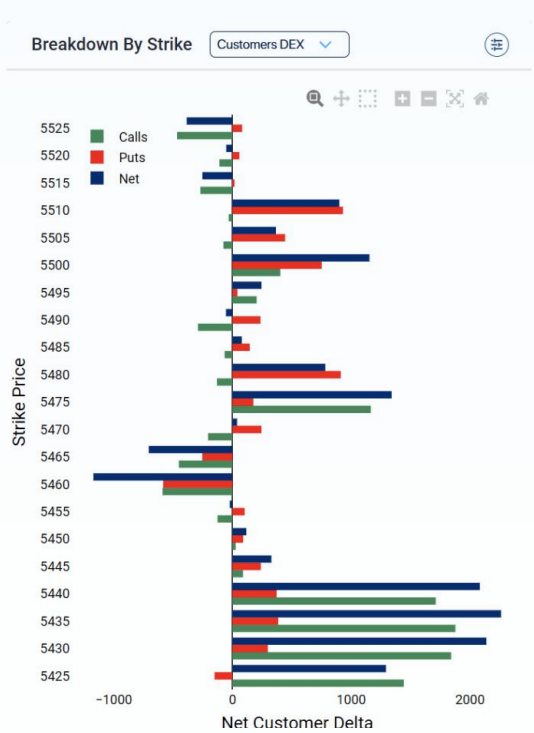
For short positions (negative position):

- The delta of a single short call ranges from -1 to 0.
- The delta of a single short put ranges from 0 to 1.
- In-the-money short options have a greater weight, with the delta of a single call approaching -1 and a single put approaching 1.

The customer DEX breakdown by strike showcases the delta exposure weight each particular contract exerts on the current overall exposure. Delta is dynamic and changes as market conditions change, but the overall directional bias remains tied to the position. For example:

- Short calls have a negative delta.
- Long calls have a positive delta.
- Short puts have a positive delta.
- Long puts have a negative delta.

The magnitude of these deltas can change significantly as market conditions change, influenced by factors such as the underlying price, time-to-expiration, and fixed strike implied volatility (IV).



Customers DEX Potential Scenarios

Long Calls have a positive delta, indicating a bullish outlook. As the underlying asset's price rises, the delta and value of the calls increase, reflecting the customers' expectation of upward price movement. This positive delta contributes to an overall bullish bias.

Long Puts have a negative delta, indicating a bearish outlook. As the underlying asset's price falls, the delta and value of the puts increase, reflecting the customers' expectation of downward price movement. This negative delta contributes to an overall bearish bias.

Short Calls have a negative delta, indicating a bearish outlook. When customers sell calls, they are betting against the price rising above the strike price. The negative delta of short calls means that as the underlying price increases, the position loses value, consistent with a bearish sentiment.

Short Puts have a positive delta, indicating a bullish outlook. When customers sell puts, they expect the price to stay above the strike price. The positive delta of short puts means that as the underlying price increases, the position gains value, reflecting a bullish sentiment.

Key Takeaway

Looking at the net Delta Exposure (DEX) can provide valuable insights into the overall bias consensus for a given strike, whether through calls or puts, long or short. This analysis helps understand the directional sentiment and potential market movements based on customer positioning.

Market Maker's Gamma Exposure (MM GEX)

Net Customer Position	Net MM Position	MM Gamma	Side	Color of the Bar
Net Long **Calls**	Net Short **Calls**	Negative(-)	Left	**Green**
Net Long **Puts**	Net Short **Puts**	Negative(-)	Left	**Red**
Net Short **Calls**	Net Long **Calls**	Positive (+)	Right	**Green**
Net Short **Puts**	Net Long **Puts**	Positive (+)	Right	**Red**

Gamma Characteristics

Gamma measures how much the delta of an option changes as the price of the underlying asset changes. Tracking gamma is particularly important for delta hedging market participants, notably market makers. They typically maintain no directional exposure and use delta hedging to neutralize their delta.

Understanding how their delta will change as price moves helps infer their buying or selling behavior as certain price levels are approached.
The MM GEX breakdown by strike showcases the gamma exposure weight each particular contract exerts on the current overall exposure. Gamma is dynamic and changes as market conditions change, but the overall polarity is tied to the position. For instance:

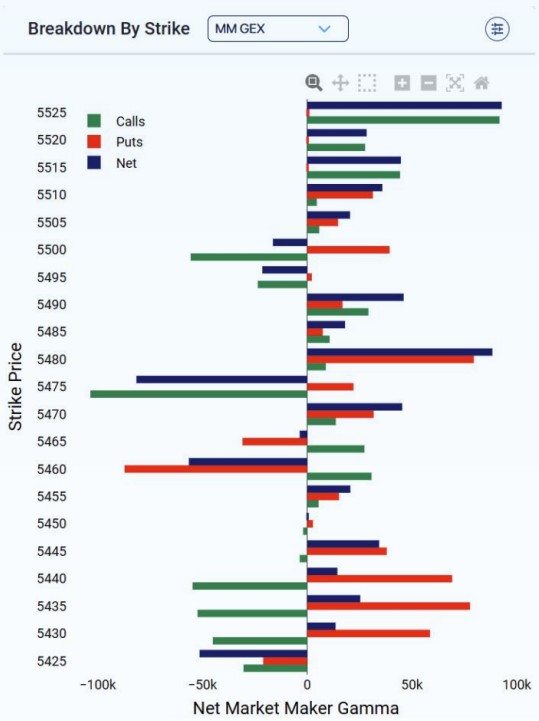
- **Long options** always have a **positive gamma.**
- **Short options** always have a **negative gamma.**
  - Gamma peaks in magnitude at the strike price.
  - Gamma gets significantly amplified as we near expiration.

The magnitude of these gamma values can change significantly due to factors such as the underlying price, time-to-expiration, and fixed strike implied volatility (IV). Refer to our heatmap, which projects upcoming changes in time and underlying price on the entire MM options portfolio.

MM GEX and Market Dynamics

Market makers hold the opposite side of customers' net positions, filling the imbalance of overall supply or demand for specific contracts. Their gamma exposure (MM GEX) is directly influenced by their overall long or short option position, regardless of the contract type. Understanding MM GEX provides insights into potential market support and resistance levels.

- **Positivie Gamma:** Market makers have a net long option position. As the underlying price moves, market makers hedge by buying when the price drops and selling when it rises. This behavior stabilizes the market, creating support on the downside and resistance on the upside, and helps dampen price volatility.
- **Negative Gamma:** Market makers have a net short gamma position. As the underlying price moves, market makers hedge by selling when the price drops and buying when it rises. This behavior exacerbates price movements, acting like a "magnet" that pulls the price further in the direction of the movement, increasing price volatility.



MM GEX Potential Scenarios

- **MM Long Options (regardless of contract type):** When customers have negative net options positions, market makers have a positive position, resulting in positive gamma exposure. They hedge against price movements, creating support and resistance levels.

- **MM Short Options (regardless of contract type):** When customers have positive net options positions, market makers have a negative position, resulting in negative gamma exposure. They need to hedge by selling as the price increases and buying as the price decreases, amplifying market movements.

Key Takeaways

- **Positive Gamma** acts as support and resistance due to market makers' hedging activities against price movements.
- **Negative Gamma** acts like a "magnet," pulling prices further in the direction of the movement, thus increasing volatility.

By analyzing the MM GEX, traders can gain insights into potential market behavior and key price levels influenced by market makers' hedging strategies. This understanding helps predict potential support and resistance zones and anticipate market volatility.

The Bottom Line

Understanding the breakdown of net positions by strike and the resulting delta and gamma exposures provides invaluable insights into market dynamics. By visualizing net positions, traders can quickly grasp the directional biases and potential hedging actions of market participants as prices approach specific levels.

This comprehensive analysis allows traders to anticipate market behavior, identify key price levels, and make more informed trading decisions. The insights gained from examining net customer positions, delta exposure, and market maker's gamma exposure equip traders with a strategic edge in the market.
We'll explore additional topics and delve deeper into other aspects of options trading in future texts. Stay tuned for more in-depth analyses and strategies to enhance your trading expertise.